(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

4th Semester, April, 2024

Paper Code: BS-202 **Subject: Probability, Statistics and Linear Programming**
Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	Two bolts are drawn from a box containing 6 good and 8 defective bolts. Find the probability that the second bolt is good if the first one is found to be defective.	2.5	CO1
1(b)	Find the value of k and mean value of X. If X is continuous random variable with probability density function given by $f(x)=kx \ (0\le x\le 2)$ $=2k \ (2\le x\le 4)$	2.5	CO1
1(c)	If X and Y are jointly distributed random variables, then prove that $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$	2.5	CO2
1(d)	What is the Central Limit theorem & why it is useful?	2.5	CO2
2(a)	A variate X has the probability distribution function x: -3 6 0 P(x): 1/6 1/2 1/3 Then find the expectation E(x) and $E(x^2)$, hence evaluate $E(2x+1)^2$.	5.0	CO1
2(b)	In a sample of 1000 cases, the mean of a certain test is 14 and deviation is 2.5. Assuming the distribution to be normal, find (i) How many students score between 12 and 15? (ii) How many score below 8? (iii) How many score above 16?	5.0	CO1
3(a)	The Join Probability distribution of two random variables X and Y is given by $P(X=0,Y=1)=1/3$, $P(X=1,Y=-1)=1/3$ and $P(X=1,Y=1)=1/3$ Find (i) the marginal distributions of X and Y. (ii) the conditional probabilities distribution of X, given by $Y=1$.	5.0	CO2
3(b)	Find the moment generating function of the exponential distribution $f(x)=\frac{1}{c}e^{-x/c}$, $0\le x\le\infty, t>0$, hence find its mean and Standard deviation if $\lvert t\rvert<\frac{1}{c}$.	5.0	CO2
4(a)	An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year? (given that $e^{-0.1}= 0.9048$)	5.0	CO1
4(b)	X is the variate with mean 30 and Standard deviation 5, then find the Probability for (i) $26\le X\le 40$ (ii) $\lvert X-30\rvert>5$	5.0	CO2

objective?

	Course 1	Course 2	Course 3	Course 4
Prof 1	60	40	60	70
Prof 2	20	60	50	70
Prof 3	20	30	40	60
TA	30	10	20	40

b) Solve the following LPP using Simplex algorithm.

Max Z = -2x +3y
subject to
$x \leq 5$
$2x-3y \leq 6$
$x, y \geq 0$

Q9. a) Find the initial basic feasible solution of the following transportation problem using Vogel's Approximation method. **[7+8=15]**

	I	II	III	IV	Supply
A	21	16	25	13	11
B	17	18	14	23	13
C	32	27	18	41	19
Demand	6	10	12	15	

b) Three food products are available at costs of Rs. 10, Rs. 36 and Rs. 24 per unit, respectively. They contain 1,000, 4,000 and 2,000 calories per unit, respectively and 200, 900 and 500 protein units per unit, respectively. It is required to find the minimum-cost diet containing at least 20,000 calories and 3,000 units of protein. Formulate and solve the given problem as an LP problem. Write the dual and using the final simplex table of primal problem, write the solution to the dual problem.

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(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-202 ***Subject: Probability, Statistics & Linear Programming***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questionin all including Q.No1 which is compulsory. Select one question from each unit. Assume missing data if any. (Scientific Calculators are allowed).

Q1. Attempt all Questions:- **3**

a) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. If two parts are selected at random and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

b) State the Central Limit Theorem. **2**

c) Find the moment-generating function of a random variable X following an exponential distribution. **3**

d) Explain the general procedure for Hypothesis Testing. **2**

e) Solve the following LPP graphically. **5**

$$Min\ Z = -3x + 2y$$
subject to
$$x + y \leq 5$$
$$4 \geq x \geq 0, 6 \geq y \geq 1$$

UNIT I

Q2. a) Let the random variable $\boldsymbol{X}$ denote the current measured in a thin copper wire in milliamperes. Assume that the range of $\boldsymbol{X}$ is [0, 20 mA], and assume that the probability density function of $\boldsymbol{X}$ is $f(x) = 0.05, 0 \leq x \leq 20$. What is the probability that a measurement of current is between 5 and 10 milliamperes? Also, determine the mean and standard deviation of $\boldsymbol{X}$. **[7+8=15]**

b) Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

i. If a new design attains a good review, then what is the probability that it will be a highly successful product?